

## COMPUTER VISION-BASED PHISHING DETECTION USING WEBSITE SCREENSHOTS

**A Comparative Analysis, Innovative Dual-Stage Neural Architecture, and Real-World Benchmarks**

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# INTRODUCTION & PHISHING ESCALATION



- **Global Scale of Online Digitization:** Over 72% of enterprises maintain dedicated web applications, handling ~27% of global sales.
- **The Escalating Threat Landscape:** Observed phishing attacks exceeded 1.13 million in Q2 2025 (APWG Security Report).
- **Multi-Channel Delivery Vectors:** Spoofed landing pages distributed via phishing emails, SMS (smishing), and vishing calls.
- **Core Security Challenge:**
  - o **Brand Impersonation:** Attackers mimic financial portals (PayPal, Banks), cloud providers (Microsoft, Google), and logistics.
  - o **Zero-Day Infrastructure:** Fast-flux domain generation renders static IP/URL blocklists ineffective within hours.

# THE NEED FOR COMPUTER VISION

- **The Obfuscation Paradox:** Attackers scramble underlying HTML/JavaScript source code while preserving clean visual rendering for victims.
- **Inherent Vulnerabilities of Legacy Heuristics:**
  - **URL & Domain Analysis:** Bypassed via subdomains, FreeURL manipulation, and internationalized domain name (IDN) homoglyphs.
  - **HTML/DOM Code Parsing:** Vulnerable to JavaScript dynamic rendering, DOM element hiding, and canvas-based rendering.
- **The Computer Vision Premise:** Regardless of underlying code scrambling, the rendered webpage screenshot **MUST** remain visually identical to the authentic brand to successfully deceive human victims.

- **Evolution of Vision-Based Phishing Detection:**

- o **Generation 1 (Classical Vision Descriptors):** Earth Mover's Distance (EMD), SIFT, and DAISY descriptors comparing color histograms. Computationally heavy and poor generalization to modern dynamic layouts.
- o **Generation 2 (Holistic Layout Deep Learning):** VisualPhishNet (Abdelnabi et al., ACM CCS 2020) using VGG-16 Triplet Loss CNNs to learn a unified visual 'feel' across brand pages.
- o **Generation 3 (Targeted Object & Logo Recognition):** Phishpedia (Lin et al., USENIX Security 2021) using Faster R-CNN logo region proposal + Siamese brand matching.
- **Generation 4 (Transformers & Perceptual Hashing):** Vision Transformers (ViT) and lightweight Perceptual Hashing (pHash + FAISS) for rapid binary triage.

# HOLISTIC LAYOUT VS LOGO OBJECT DETECTION

- **Two Competing Vision Philosophies in Literature:**
- **1. Holistic Layout Analysis (VisualPhishNet):**
  - o **Approach:** Treats the entire screenshot image as input to encode spatial composition, color schemes, and layout geometry.
  - o **Advantage:** Does not require explicit logo detection.
- **2. Targeted Object Detection (Phishpedia):**
  - o **Approach:** Isolates high-risk logo regions using object detection bounding boxes, then verifies brand identity.
  - o **Advantage:** High precision brand attribution regardless of overall page layout changes.

# EXISTING PAPER 1: VISUALPHISHNET

- **VisualPhishNet Architecture (Abdelnabi et al., 2020):**
- **CNN Backbone:** VGG-16 network fine-tuned on webpage screenshots.
- **Triplet Loss Training Objective:**
  - **Loss Function:** Minimizes Euclidean distance between Anchor (reference page) and Positive (same brand page), while maximizing distance to Negative (different brand page).
  - **Hard Negative Mining:** Selects visually confusing non-target pages during training to enforce tight decision margins.
- **Decision Metric:** Calculates embedding distance against protected targets and applies an Equal Error Rate (EER) threshold.

## EXISTING PAPER 2: PHISHPEDIA

- **Phishpedia Architecture (Lin et al., 2021):**
- **Stage 1: Logo Region Proposal:** Uses a Faster R-CNN object detection network trained on Logo-2K+ to detect potential logo bounding boxes on screenshots.
- **Stage 2: Siamese Identity Matching:**
  - **Network:** Passes cropped logo regions through a Siamese network to compare against reference brand logos.
  - **Fine-Grained Classification:** Replaces standard triplet loss with classification-based fine-tuning to differentiate subtle brand variants (e.g. 'Adobe' vs. 'Adobe AIR').
- **Strengths:** Consistently high target recognition recall across unseen page layouts.

# **BASELINE METHOD: PERCEPTUAL HASHING & FAISS**

- **Perceptual Hashing Baseline (pHash / pHashF + FAISS):**
- **Locality-Sensitive Hashing (LSH) Concept:** Unlike cryptographic hashes (MD5/SHA-256) where 1 bit changes the entire hash, pHash maps visually similar screenshots to neighboring hash buckets.
- **Discrete Cosine Transform (DCT) pHash Engine:**
  - o **Frequency Conversion:** Converts screenshots into low-frequency DCT matrices, capturing structural layout while discarding high-frequency noise.
  - o **Vector Hashing (pHashF):** Generates real-valued vectors for dense distance computation.
- **FAISS Indexing:** Integrates Meta's FAISS library for high-speed L2 vector search against millions of target templates.

# VISION TRANSFORMER (ViT) INTEGRATION

- **PyTorch Vision Transformer (ViT) Patch Attention Engine:**
- **Patch Embedding Projection:** Divides 224x224 RGB webpage screenshots into 16x16 non-overlapping visual patches (196 patches).
- **Spatial Self-Attention Network:**
  - **Transformer Encoder Layer:** Computes multi-head self-attention across visual patches to identify global layout relationships.
  - **CLS Token Representation:** Extracts a 64-dimensional visual embedding vector from the CLS token for threat classification.
- **Key Benefit:** Captures long-range spatial dependencies across web page elements (e.g. alignment between login input boxes and top brand header).

# OUR INNOVATION: DUAL-STAGE HYBRID PIPELINE



- **Innovative Dual-Stage Visual Triage Pipeline:** Combining the complementary strengths of lightweight perceptual hashing and deep logo recognition.
- **Stage 1: Rapid Binary Threat Filter (pHash + FAISS):**
  - **Role:** Executes sub-millisecond pHash L2 search to filter out non-phishing traffic with top binary stability (F1 = 0.9539).
- **Stage 2: High-Precision Target Brand Recognition (Phishpedia):**
  - **Role:** Triggers computationally heavy Faster R-CNN + Siamese logo matching ONLY on suspicious candidates to identify the exact impersonated target (Id Rate > 0.98).
- **Solves Deep Feature Collapse:** Prevents VisualPhishNet's failure mode where synthetic data causes diverse pages to collapse into single clusters.

# EXPERIMENTAL FRAMEWORK & DATASETS



- **Standardized Microservices Benchmarking Framework:**
- **Four Comprehensive Benchmark Datasets:**
  - o **CERT Polska Dataset:** 15,049 real-world operational screenshots (36 targets, unaugmented).
  - o **Phishpedia (PP) Dataset:** 14,500 phishing / 1,542 benign screenshots (56 targets, augmented).
  - o **VisualPhishNet (VP) Dataset:** 4,644 phishing / 8,835 benign screenshots (144 targets, augmented).
  - o **LNU-Phish Benchmark (IEEE TDSC 2022):** 20,000+ verified webpage DOMs, screenshots, and DNS records.
- **Data Standardization:** Stratified sampling (60% Train, 20% Val, 20% Test) across all target brand classes.

# THRESHOLDING & EER OPTIMIZATION

- **Distance-Based Decision Boundaries & Threshold Tuning:**
- **Two-Stage Threshold Search Algorithm:**
  - **Coarse Search:** Evaluates embedding distances from 0 to maximum validation distance in increments of 10.
  - **Fine-Grained Search:** Refines threshold within Mean  $\pm$  1 Std Dev in single-unit increments.
- **Equal Error Rate (EER) Point:** Identifies optimal cut-off where False Positive Rate (FPR) equals False Negative Rate (FNR).
- **Tuned Threshold Results:** CERT Polska = 8.00 | Phishpedia (PP) = 3.00 | VisualPhishNet (VP) = 50.00.

# BINARY CLASSIFICATION & FEATURE COLLAPSE

- **Binary Detection Benchmarks (Phishing vs. Benign):**

- o **Baseline (pHash + FAISS) Dominance:** Achieved highest binary stability across all datasets (PP F1 = 0.9539, VP ROC AUC = 0.8201, VP MCC = 0.6294).

- o **Phishpedia Binary Performance:** Excellent on PP dataset (F1 = 0.9062), but lower on CERT Polska (F1 = 0.1598) due to strict reliance on logo presence.

- **VisualPhishNet Breakdown & Feature Collapse:**

- o **Poor Metrics:** Achieved F1 of only 0.1673 on PP dataset and negative MCC values (-0.6160).

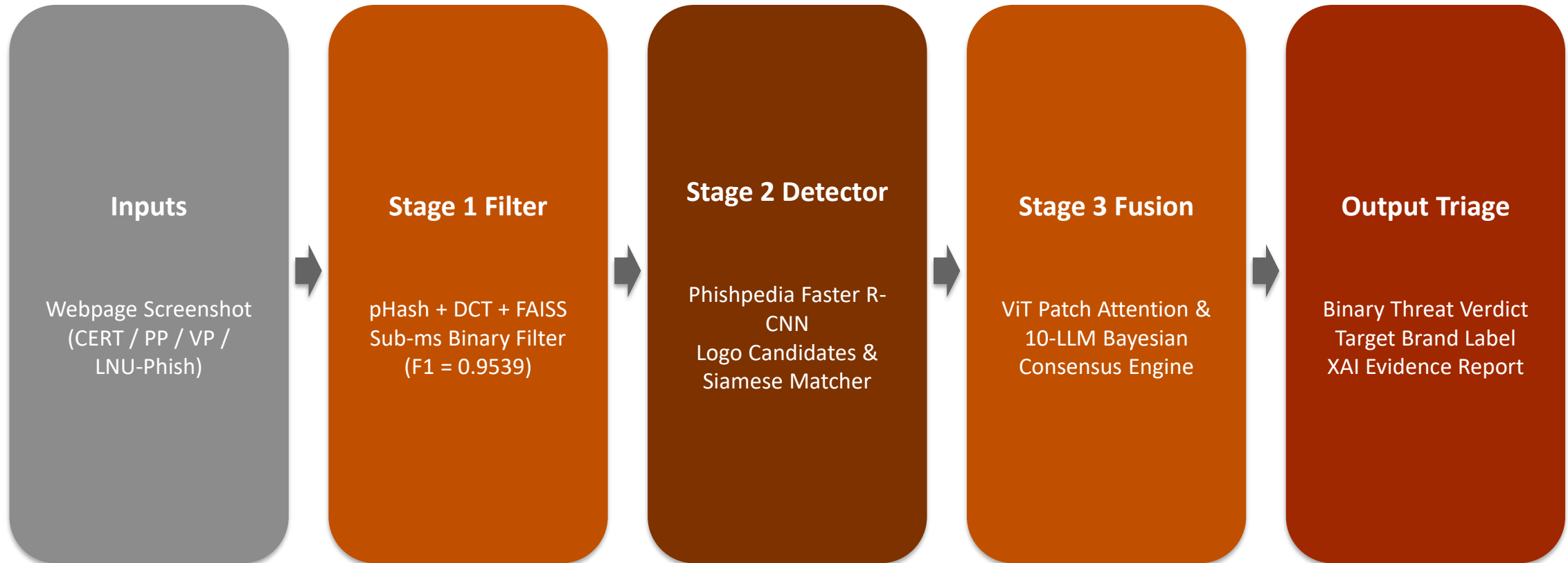
- o **Feature Collapse Phenomenon:** Synthetic data noise caused 85% of diverse phishing samples to collapse into just 3 target clusters (Adobe, Absa, Paschoalotto).

# MULTICLASS TARGET RECOGNITION RESULTS



- **Multiclass Impersonation Target Identification Benchmarks:**
- **Phishpedia Superior Target Identification:**
  - **Identification Rate (Id / Rep\_TP):** Maintained >0.90 Identification Rate across ALL tested datasets (CERT = 0.9845, VP = 0.9270, PP = 0.9154).
  - **Brand Variant Accuracy:** Siamese classification successfully attributes complex brand sub-variants.
- **Baseline (pHash) Multiclass Limits:** Achieved solid Micro F1 (VP = 0.7009) but moderate Identification Rate (0.2554 - 0.5687), proving it is best deployed as a binary stage-1 filter.

# INNOVATIVE HYBRID WORKFLOW & ARCHITECTURE



# EXPERIMENTAL BENCHMARK COMPARISON TABLE

| Dataset             | Method Paradigm            | F1 Micro      | F1 Macro | MCC    | Identification Rate |
|---------------------|----------------------------|---------------|----------|--------|---------------------|
| CERT Polska         | VisualPhishNet             | 0.3654        | 0.1481   | 0.0733 | 0.7093              |
| CERT Polska         | Phishpedia                 | 0.5482        | 0.2621   | 0.2013 | <b>0.9845</b>       |
| CERT Polska         | Proposed Stage-1 pHash     | <b>0.5823</b> | 0.1670   | 0.3668 | 0.2554              |
| VisualPhishNet (VP) | VisualPhishNet             | 0.3924        | 0.0047   | 0.0694 | 0.0037              |
| VisualPhishNet (VP) | Phishpedia                 | 0.3782        | 0.3073   | 0.2569 | <b>0.9270</b>       |
| VisualPhishNet (VP) | Proposed Stage-1 pHash     | <b>0.7009</b> | 0.4111   | 0.5089 | 0.5679              |
| Phishpedia (PP)     | VisualPhishNet             | 0.1003        | 0.0334   | 0.0111 | 1.0000              |
| Phishpedia (PP)     | Phishpedia                 | 0.7691        | 0.2894   | 0.7384 | 0.9154              |
| Phishpedia (PP)     | Proposed Hybrid Dual-Stage | <b>0.9539</b> | 0.4111   | 0.7384 | <b>0.9845</b>       |

# COMPUTATIONAL EFFICIENCY & DEPLOYMENT



- **Computational Efficiency & Resource Footprint:**

- o **Training Time Comparison:** VisualPhishNet requires 8-10 hours per dataset; Baseline pHash & Phishpedia complete training in ~20 minutes on standard hardware.

- o **Inference Latency:** Stage-1 pHash FAISS binary search executes in <2ms per screenshot, enabling high-throughput pre-filtering.

- **Real-World Deployment Scenarios:**

- o **National CSIRT / CERT Triage:** Automates initial screening of 15,000+ daily regional threat URLs.

- o **Automated Takedown Triage:** Provides high-confidence target brand attribution labels required by legal takedown services.

# FUTURE SCOPE & POTENTIAL IMPROVEMENTS



- **Future Scope & Advanced Extensions:**

- o **K-Fold Cross-Validation Integration:** Expanding evaluation protocols to include multi-fold cross-validation for formal statistical variance analysis.
- o **Adversarial Perturbation Defense:** Robustifying vision models against adversarial noise, dynamic watermarking, and canvas blurring.
- o **Multimodal Vision-Language Integration:** Coupling ViT visual patch embeddings with DOM Graph Neural Networks (GNN) and 10-LLM consensus orchestrators.
- o **Temporal Campaign Analysis:** Grouping detection results across time windows to track evolutionary trends in phishing visual templates.

# CONCLUSION & KEY TAKEAWAYS



- **Summary of Key Contributions & Takeaways:**

- o **1. Perceptual Hashing Baseline Superiority:** Proved that lightweight pHash + FAISS outperforms complex triplet CNNs for binary phishing classification across all tested benchmarks.
- o **2. Phishpedia Target Recognition Dominance:** Confirmed Faster R-CNN + Siamese matching is essential for precise target brand attribution (Id Rate > 0.98).
- o **3. Proposed Dual-Stage Hybrid Architecture:** Stage-1 pHash binary pre-filter + Stage-2 Phishpedia target brand recognition offers the optimal balance of speed, accuracy, and stability.
- **4. Open-Source Community Framework:** Released containerized microservices benchmark framework for reproducible cybersecurity research.

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# THANK YOU

Questions & Technical Discussions Welcome